

# TASK 2 — Case Study: Drug Formulation

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## Tablet Formulation Case Study — Paracetamol Immediate Release Tablet

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### Paracetamol Immediate Release Tablet Formulation

#### Pharmaceutical Formulation Case Study

Prepared for: CodeAlpha Pharmaceutical Research Internship

Topics Covered:

- Formulation process
  - Excipients used
  - Manufacturing techniques
  - Stability factors
  - Industrial challenges
  - Quality control
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## Introduction to Paracetamol Tablets

### About Paracetamol

Paracetamol (Acetaminophen) is one of the most widely used analgesic and antipyretic drugs globally.

#### Therapeutic Uses

- Fever reduction
- Mild to moderate pain relief
- Post-operative pain management
- Headache and muscle pain

#### Dosage Form Chosen

Immediate Release Oral Tablet

Advantages

- Easy administration
- Rapid onset of action
- Cost effective
- High patient compliance

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Formulation Components

Major Excipients Used

| Excipient                  | Function           |
|----------------------------|--------------------|
| Microcrystalline cellulose | Binder and filler  |
| Pregelatinized starch      | Disintegrant       |
| Povidone K30               | Binder             |
| Magnesium stearate         | Lubricant          |
| Talc                       | Glidant            |
| Sodium starch glycolate    | Super disintegrant |
| Colloidal silicon dioxide  | Flow enhancer      |

API

Paracetamol — Active pharmaceutical ingredient responsible for therapeutic action.

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Manufacturing Process

Wet Granulation Method

Step-by-Step Process

1. Weighing and dispensing
2. Dry mixing of API and excipients
3. Binder solution preparation
4. Wet granulation
5. Drying process

6. Milling and sizing
7. Lubrication blending
8. Tablet compression
9. Coating (optional)
10. Packaging

### Process Advantages

- Better flow properties
  - Improved compressibility
  - Uniform drug distribution
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## Stability Factors

### Critical Stability Considerations

| Factor               | Impact               |
|----------------------|----------------------|
| Temperature          | Degradation risk     |
| Moisture             | Tablet softening     |
| Light exposure       | Chemical instability |
| Oxidation            | Reduced potency      |
| Compression pressure | Hardness variation   |

### Stability Testing

- Accelerated stability studies
- Long-term stability studies
- Dissolution testing
- Content uniformity testing

### Packaging Solutions

- Blister packs
  - HDPE containers
  - Moisture-resistant packaging
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## Formulation Challenges

# Industrial Challenges

## Common Problems

- Poor powder flow
- Sticking during compression
- Capping and lamination
- Moisture sensitivity
- Dose uniformity issues

## Solutions

- Optimized granulation
- Lubricant adjustment
- Controlled humidity
- Improved particle size distribution
- Advanced process monitoring

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# Quality Control and Evaluation

## Evaluation Parameters

| Test                | Purpose              |
|---------------------|----------------------|
| Hardness test       | Mechanical strength  |
| Friability test     | Durability           |
| Dissolution test    | Drug release         |
| Weight variation    | Dose consistency     |
| Assay testing       | Potency verification |
| Disintegration test | Release speed        |

## GMP Importance

Good Manufacturing Practices ensure product quality, safety, and regulatory compliance.

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## Conclusion:

Paracetamol tablet formulation demonstrates the importance of excipient compatibility, manufacturing optimization, and stability management in pharmaceutical development.

The formulation process requires careful balance between:

- Drug efficacy
- Manufacturing efficiency
- Patient safety
- Stability performance
- Regulatory compliance

Advanced pharmaceutical manufacturing technologies continue to improve product quality, scalability, and therapeutic effectiveness.